

Project Title: AI-Powered Smart Wheelchair for Disabled Individuals with Voice Control

Team no:T10

Team members Name with USN (use CAPITAL letters)

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1. Introduction

AI-based Smart and Intelligent Wheelchair

Published In / Year: Journal of Applied Research and Technology, 2020

Link: [Source]

Key Contributions / Findings: Proposed a wheelchair integrating AI for autonomous navigation and health monitoring, enabling user self-dependency.

Limitations Identified: Limited real-world testing and adaptability to diverse environments.

Relevance to Your Project: Demonstrates the feasibility of combining AI with health monitoring in smart wheelchairs.

Research Gap Identified: Need for more robust, adaptive AI models for varied user needs and settings.

2. Literature Survey

1. AI-based Smart and Intelligent Wheelchair

Published In / Year: Journal of Applied Research and Technology, 2020

Link: Journal of Applied Research and Technology, 2020

[pdf source](#)

Limitations Identified: Dependency on clear voice input; limited to simple commands.

Relevance to Your Project: Validates the use of voice recognition for hands-free wheelchair operation.

Research Gap Identified: Need for advanced voice processing and multi-modal controls.

2).AI Powered Smart Wheelchair

Published In / Year: IJSDR, 2020

Link: IJSDR, 2020

[pdf source Link](#)

Key Contributions / Findings: Developed a voice-controlled wheelchair using NodeMCU and Android app, improving ease of use for users with severe disabilities.

Limitations Identified: Dependency on clear voice input; limited to simple commands.

Relevance to Your Project: Validates the use of voice recognition for hands-free wheelchair operation.

Research Gap Identified: Need for advanced voice processing and multi-modal controls.

3)AI Advances in Wheelchair Navigation and Control: A Comprehensive Review

Published In / Year: Journal of Process Management and New Technologies, 2023

Link: [pdf source Link](#)

Journal of Process Management and New Technologies, Vol. 11, Issue 3-4, 2023

Key Contributions / Findings: Systematic review of AI techniques (computer vision, ML, path planning) for navigation, safety, personalization, and predictive maintenance in wheelchairs.

Limitations Identified: Challenges in real-time responsiveness and adaptation to new environments.

Relevance to Your Project: Provides a broad overview of AI applications and their benefits in wheelchair technology.

Research Gap Identified: Integration of robust, real-time AI systems for dynamic and unpredictable environments.

4)Using Artificial Intelligence to Share Control of a Powered-Wheelchair

Published In / Year: University of Portsmouth, 2018

Link: University of Portsmouth, 2018

[Source](#)

Key Contributions / Findings: Explored shared control between user and AI, improving navigation and safety.

Limitations Identified: User adaptation to shared control can be challenging.

Relevance to Your Project: Highlights the importance of user-AI collaboration for optimal control.

Research Gap Identified: Balancing autonomy and user control in real-world scenarios.

5) AI-powered Navigation for Wheelchairs

Published In / Year: Journal of Technology Innovations and Energy

Link: [pdf source Link](#)

Key Contributions / Findings: Implemented AI-powered navigation using computer vision for obstacle avoidance.

Limitations Identified: Limited performance in low-light or cluttered environments.

Relevance to

Your Project: Supports the use of AI for safe, autonomous movement.

Research Gap Identified: Need for improved sensor fusion and environmental adaptability.

6) Smart Wheelchair with Voice and Gesture Control

Published In / Year: IJETER

Link: International Journal of Emerging Technologies in Engineering Research (IJETER)

[pdf source](#)

Limitations Identified: System complexity and potential for misinterpretation of commands.

Relevance to Your Project: Demonstrates the value of multi-modal interfaces.

Research Gap Identified: Simplifying user interface without sacrificing functionality.

7. Deep Reinforcement Learning for Personalized Wheelchair Control

Published In / Year: Year: University of Portsmouth News, 2023

Link: [scholar citation Link](#)

Key Contributions / Findings: Used deep RL to optimize navigation and personalize control for individual users.

Limitations Identified: High computational requirements; limited real-time deployment.

Relevance to Your Project: Shows how AI can adapt to user preferences and abilities.

Research Gap Identified: Efficient real-time implementation on embedded systems.

8. Computer Vision-Based Obstacle Detection in Wheelchairs

Published In / Year: Cited in AI Advances in Wheelchair Navigation and Control, 2023

Link:[Source link](#)

Key Contributions / Findings: Applied computer vision for real-time obstacle detection, improving user safety.

Limitations Identified: Sensor reliability and processing speed in dynamic environments.

Relevance to Your Project: Underlines the importance of robust obstacle detection for safety.

Research Gap Identified: Enhancing accuracy and speed of detection in varied settings.

3. Problem Statement

Many individuals with severe physical disabilities face limitations in independently operating traditional or even powered wheelchairs. Current solutions often lack intuitive, adaptive, and safe control mechanisms, especially for users unable to use joysticks or manual controls. There is a need for a smart wheelchair system that leverages AI and voice recognition to provide reliable, hands-free, and adaptive mobility, ensuring safety and personalization for diverse user needs.

4.Objectives of the Project

Objective 1: Develop a smart wheelchair system with robust voice control for hands-free operation.

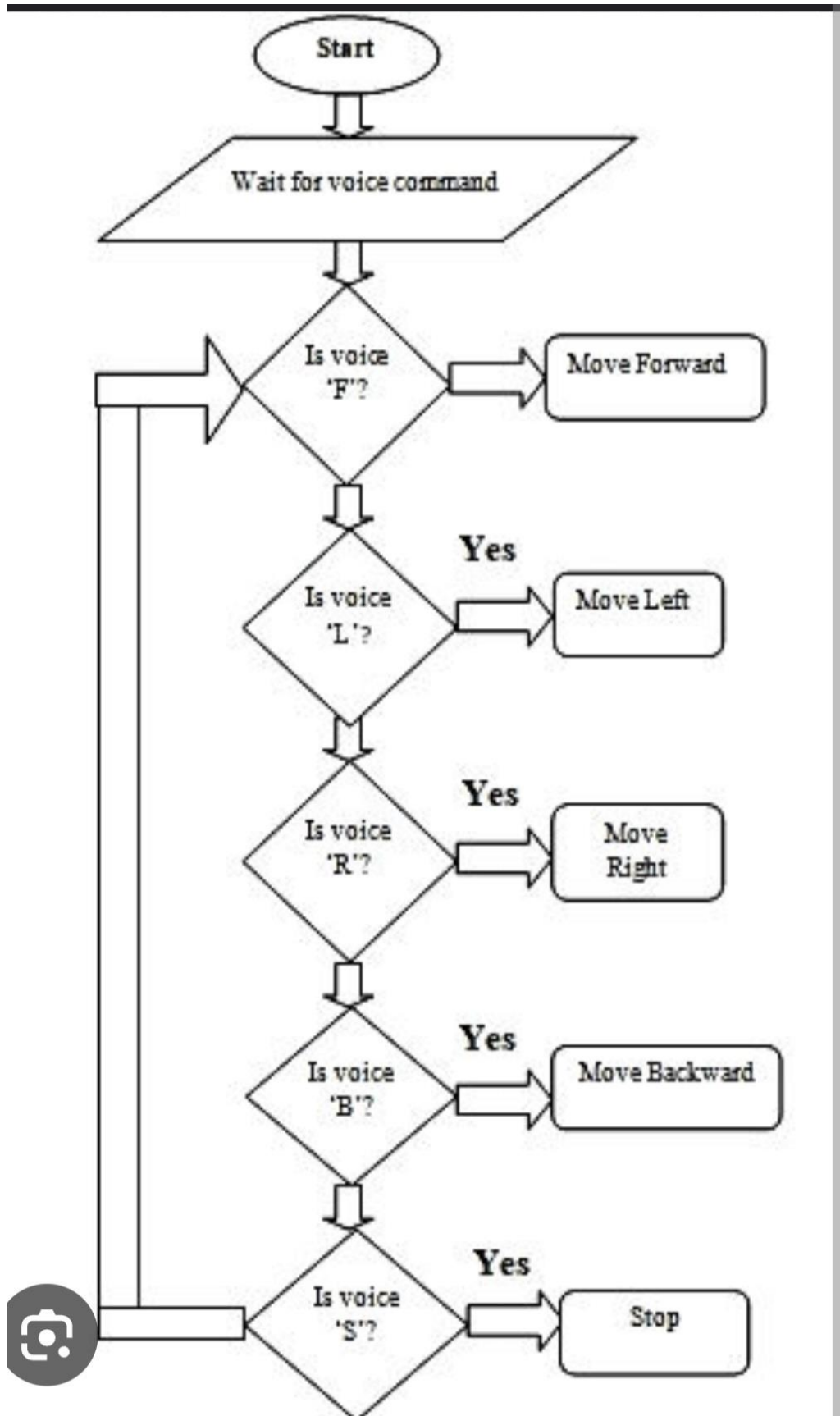
Objective 2: Integrate AI-based navigation and obstacle avoidance for safe, autonomous movement.

Objective 3: Personalize wheelchair control based on user preferences and environmental conditions.

5.System Design

5.1 Architecture Diagram

Voice Input → Voice Recognition Module → AI Controller → Motor Driver & Sensors → Wheelchair Movement & Feedback



5.2 Module Description

Voice Recognition Module: Captures and processes user commands for movement.

AI Controller: Interprets commands, plans navigation, and manages obstacle avoidance.

Motor Driver: Executes movement instructions from the AI controller.

Obstacle Detection Sensors: Continuously scan for and respond to environmental hazards.

User Feedback Interface: Provides audio or visual feedback on system status.

5.3 Flow Diagram

This literature survey and design overview provide a foundation for developing an advanced, AI-powered smart wheelchair with voice control, addressing key gaps in current solutions and focusing on user independence and safety.